Matthias Paterson-Hawes

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I'm a second year Mechanical Engineering student at the University of Auckland with a strong interest in design and manufacturing. I particularly enjoy 3D printing and rapid prototyping, using quick design iterations to test ideas and refine practical solutions.

EDUCATION

University of Auckland
BE(Hons), Mechanical Engineering — *March. 2025 – Present*

PROJECTS

Quadruped Robot Dog
Personal work

- Built a four-legged robot with hip and knee joints on each leg, using 3D-printed components and sensor inputs to control its movement and behaviour.
- It also has a human interface, with touch, sound, and PIR sensors to mimic a dog as closely as possible. It uses an ESP32-C3 and firmware written in C.
- This was 3D printed and extensively tested until it became a working prototype that walked around.

Warman group project
Group work

- The design was to deliver 2 balls at varying heights over an obstacle into a small hole, limited to an Arduino microcontroller and 4 DC motors.
- My key role was in electronics and software, with material fabrication as well, using a 3D printer to enable faster prototyping.
- My outcome was successfully making a smaller PCB for our robot with all the necessary components on a single board, writing all the fundamental functions in C and testing different 3D printing materials to minimise weight while maximising strength.
- Programmed concurrent motor operation within current limits, reducing task completion time.

RC Car
Personal work

- Built an RC car prototype using 3D-printed components, with a working differential and servo steering; tested at 55 km/h.

EXPERIENCE

My Tuition — Mar. 2026 – Present
Calculus and physics scholarship tutor

- Tutor three students in calculus and physics, supporting work towards Excellence-level achievement and teaching Scholarship Physics content.
- Deliver tutoring in person and online.

Computer Solutions Now — Feb. 2022 – Jun. 2025
Owner, Computer Repair and Assembly

- Ran a computer repair, upgrade and assembly business.
- Repaired and upgraded customers' computers, with a focus on affordable solutions.
- Built and sold desktop computers.

Caltex, Constellation Drive — Sep. 2022 – Mar. 2026
Cashier

- Provided customer service and handled cash transactions.
- Resolved customer issues and managed stocktakes.

SKILLS

CAD: SolidWorks, Fusion 360, Inventor
Programming: Python, MATLAB, Simulink, C, C++
Fabrication: 3D printing, Welding, Turning, Milling
Electronics: ESP32, Arduino, Sensor work, soldering

AWARDS AND ACTIVITIES

- Outstanding NZQA Technology Scholarship
- Level 3 Excellence Endorsement
- Participant in Aeronautics, Rocketry, and MECHA clubs